STITCH

Robust Appearance-Invariant Visual Place Recognition for Anchoring Social AR Content in the Real World

Summer Internship Presentation



PRONOB BHATTACHARJEE

25565017

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Mehta Family School of Data Science and Artificial Intelligence

Under the guidance of

Dr. R. Balasubramanian

Professor (HAG) and Head

Department of Computer Science and Engineering

Joint Faculty

Mehta Family School of Data Science and Artificial Intelligence

Presentation Outline

1. Problem and Existing System
2. Proposed Pipeline
3. Data and Experimental Protocol
4. Results and Ablations
5. Deployment and Limitations
6. Conclusions
7. References

Stitch: Social AR in the Real World

- Post digital content at a physical place.
- A later visitor points a camera at the scene.
- Recognise the place, verify its geometry and retrieve its posts.
- Render an image attached to the stored reference plane.

Research question

Can learned representations improve appearance robustness while keeping anchoring geometrically supported?

Implemented scope: local place database and planar image posts; multi-user 3D persistence is future work.

Why Appearance Change Matters

- Night changes contrast, visible detail and light-source structure.
- Viewpoint changes overlap even under unchanged daylight.
- A similar-looking place does not establish an anchor.
- Retrieval, geometric acceptance and alignment accuracy need separate metrics.

Key distinction

Correct place identity $\neq$ accurate homography $\neq$ metric camera pose.

Original Stitch Pipeline

- SIFT detection and descriptors [1].
- FLANN matching, ratio filtering and RANSAC homography.
- Perspective warp of the bundled image overlay.
- Latest script adds optical flow, detection scheduling and smoothing.

Reproduction

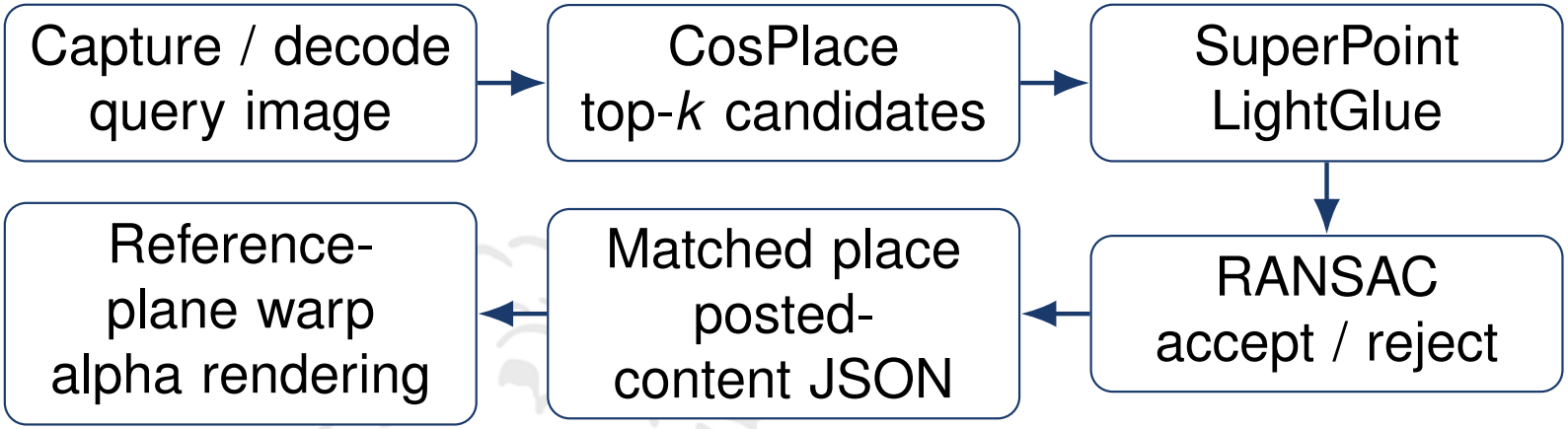
All three original scripts ran on the same bounded video input. Originals remain unchanged; a file adapter handles the unavailable private stream.

Four Methods Compared

	Method	Role
A	Classical	SIFT + FLANN + shared geometric verification
B	Learned local	SuperPoint + LightGlue, exhaustive reference matching
C	Global retrieval	CosPlace ResNet18, 512-dimensional descriptors
D	Full pipeline	Global shortlist $\rightarrow$ learned local matching $\rightarrow$ geometry $\rightarrow$ posted-content rendering

Pretrained components: SuperPoint [2], LightGlue [3], CosPlace [4].

System Architecture



Failure handling

No accepted candidate $\Rightarrow$ no anchored post is rendered.

Coarse-to-fine localisation principle: [7]; executed prototype uses planar verification.

Geometry and Posted Content

- Estimate a reference-to-query homography using tentative correspondences.
- Check inlier count, inlier ratio and projective validity.
- Retrieve content from the verified place record.
- Use an explicit alpha mask; black asset pixels remain valid content.

$$\widetilde{\mathbf{x}}_q \sim H_{r\to q}\widetilde{\mathbf{x}}_r, \qquad H_{c\to q} = H_{r\to q}A_{c\to r}.$$

$A_{c\to r}$ maps asset corners to the stored reference polygon.

Actual Evaluation Data

- HPatches: 144 images, 120 ground-truth homography pairs [5].
- Tokyo: 96 real photographs from 24 selected sites [6].
- Synthetic stress test: 90 queries from six source images and three seeds.
- Download manifests, CRC checks, SHA256 values and frozen protocols are retained.

Scope

Tokyo is a custom cross-time query-to-query subset, not the official street-view retrieval benchmark.

Real Day, Night and Sunset Images



Actual downloaded Tokyo photographs. Front/day, left/day, front/night and front/sunset metadata define each retained site.

Synthetic Appearance Stressors



Executed gamma, gain, colour shift, noise and fog-like transformations; these are not real seasonal/weather observations.

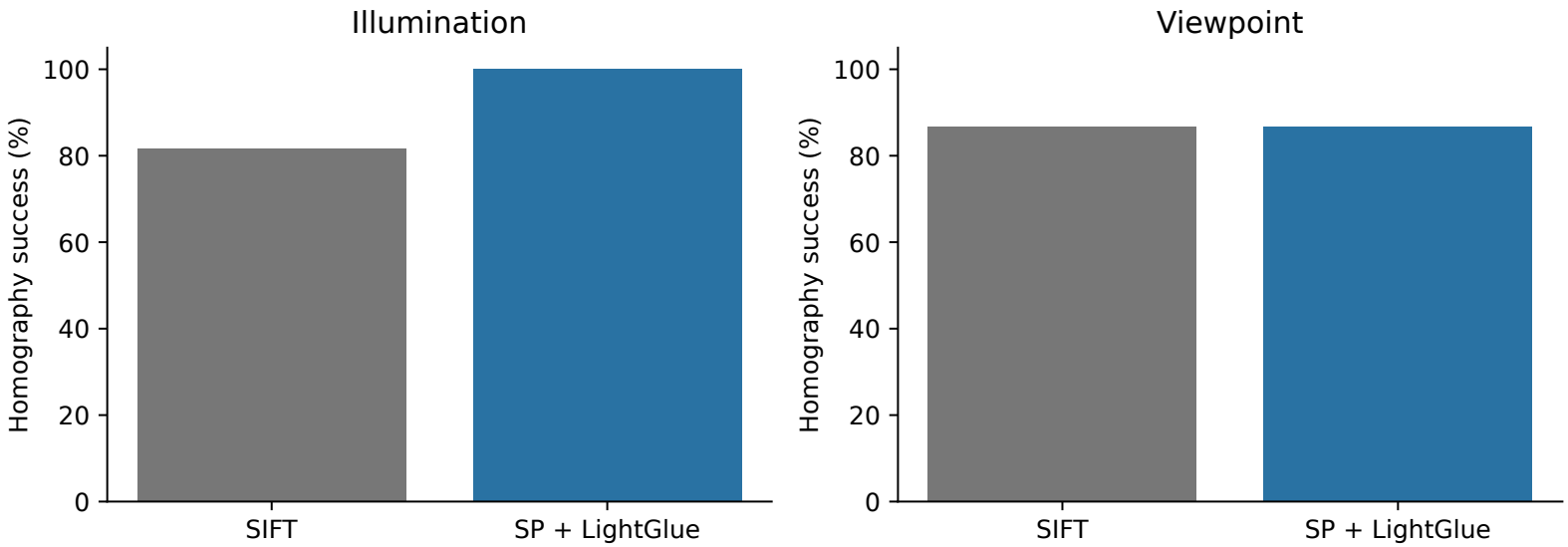
Experimental Controls and Metrics

- Same queries, reference galleries and geometric thresholds for A–D.
- Maximum image side: 640 pixels; requested keypoints: 512.
- RANSAC: 5-pixel threshold; acceptance: 12 inliers and 0.20 ratio.
- Retrieval: Recall@1/5/10; decisions: correctly verified places.
- HPatches/synthetic: accepted mean corner error ≤ 5 pixels.
- Timing: isolated CPU queries with cached references; memory: process peak RSS.

Host: Apple M2, 8 GB RAM, two CPU threads. No phone or calibrated metric-pose result is claimed.

Local Matching: HPatches

HPatches split	SIFT	SuperPoint + LightGlue
Illumination	81.7	100.0
Viewpoint	86.7	86.7



Percent successful oracle-pair homographies. Learned matching improves this illumination subset; viewpoint success is tied.

Tokyo Day-to-Night: Measured Results

Method	R@1	R@5	Verified
SIFT + FLANN	58.3	91.7	33.3
SuperPoint + LightGlue	100.0	100.0	100.0
CosPlace only	100.0	100.0	–
Full pipeline ($k = 3$)	100.0	100.0	100.0

Interpretation

The full pipeline verifies 100.0% of the selected queries. The gallery has only 24 sites and similar front-facing directions.

Rates are percentages. Global retrieval alone does not produce an anchored transform.

Viewpoint Change: A Negative Result

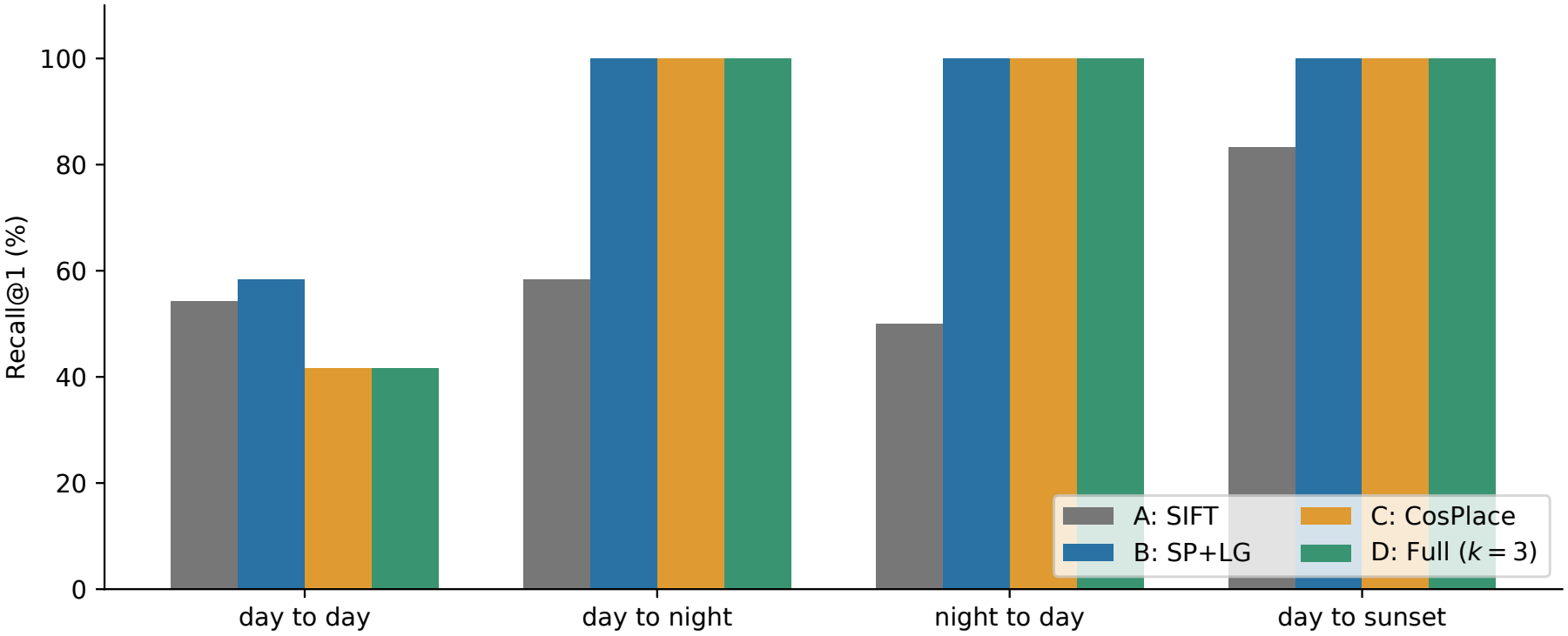
Method	R@1	R@5	Verified
SIFT + FLANN	54.2	66.7	41.7
SuperPoint + LightGlue	58.3	70.8	54.2
CosPlace only	41.7	62.5	–
Full pipeline ($k = 3$)	41.7	62.5	37.5

Global-shortlist limitation

The full pipeline correctly verifies only 37.5% when query direction changes. Exhaustive local search recovers some missed references.

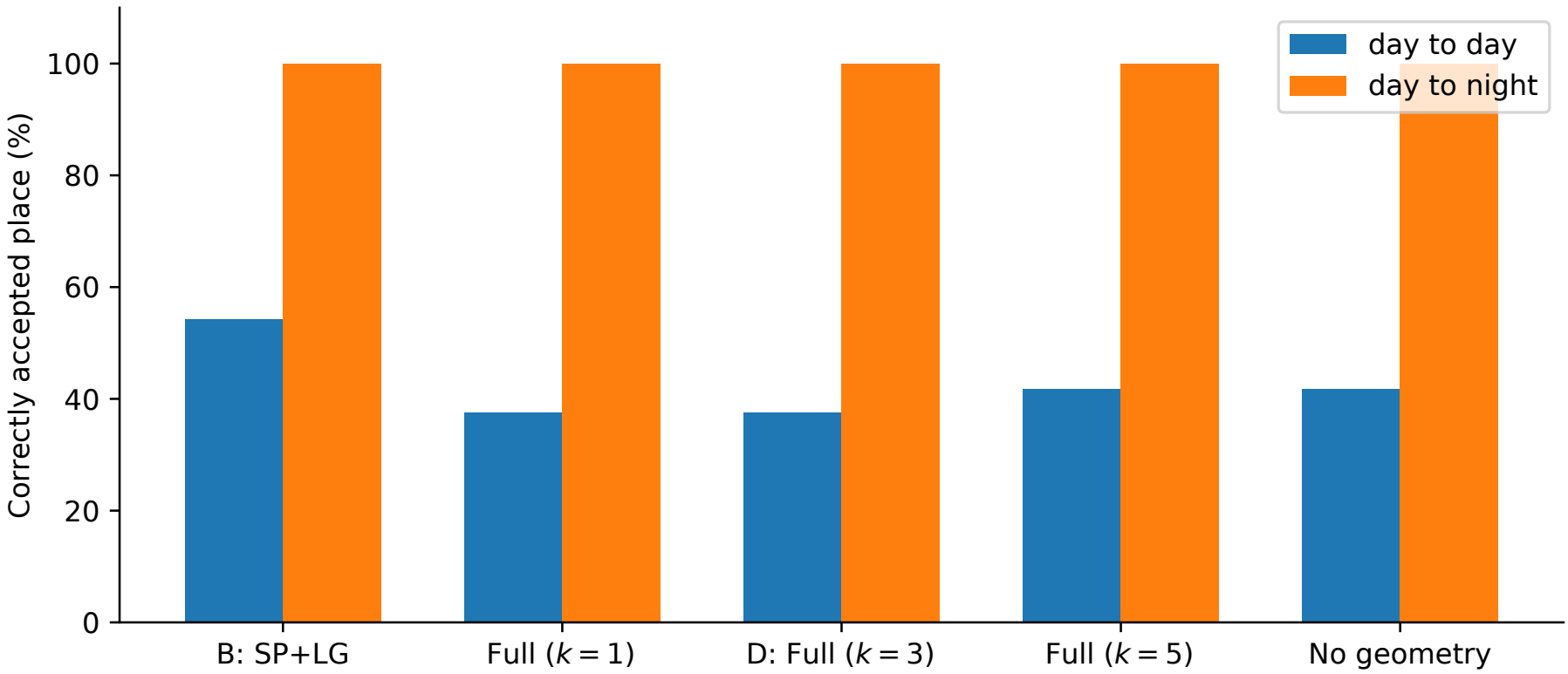
Day-to-day means front-facing daytime reference to left-facing daytime query. It is not an unchanged-view control.

Retrieval Across Real Tokyo Splits



Actual Recall@1. Similar viewpoint across time can be easier than different viewpoint in daylight.

Shortlist and Verification Ablations



Same queries: exhaustive learned search, $k = 1/3/5$, and unverified top-result acceptance. More candidates can introduce competing hypotheses.

Unknown Places and False Acceptance

Variant	False accepts	Unknowns
SIFT + FLANN	0	12
SuperPoint + LightGlue	2	12
Full pipeline ($k = 3$)	0	12
No geometry	12	12

Geometric support is not identity assurance

Some unrelated queries still pass local verification. A production AR service needs stronger negative testing and calibrated decision policies.

Disjoint-collection diagnostic: HPatches queries against a Tokyo gallery; not a production false-positive bound.

Short Adaptation: Three Fixed Seeds

Held-out split	Pretrained	Head	Head + aug.
HP viewpoint	77.5	75.8	74.2
Tokyo view	41.7	30.6	37.5
Tokyo night	100.0	94.4	94.4

- Frozen encoder; only a 512-to-128 linear projection is trained.
- Eight training and sixteen held-out HPatches places; no Tokyo training.
- With/without synthetic night augmentation; 120 steps per seed.
- No consistent improvement over the pretrained descriptor.

Mean Recall@1 (percent). The report includes seed standard deviations and intervals; compression and adaptation are not separated.

Actual Correspondences Across Day/Night

sift: 13 matches, 5 inliers; rejected



superpoint: 211 matches, 163 inliers; accepted



Recorded tentative matches: green denotes RANSAC inliers. Tokyo provides no independent homography or metric-pose ground truth here.

Anchored-Content Comparison Demo



Actual classical/full executions. Real still images with labelled generated 2D camera motion; playback speed is not inference throughput.

Measured CPU Latency and Memory

Method	Median ms	Peak RSS MB
Classical	186.2	568.3
Exhaustive learned	3020.4	1435.5
Retrieval only	18.5	537.2
Full pipeline	1332.5	1492.3

Mobile implication

Full-pipeline median: 1332.5 ms per query on this CPU host. Use occasional recognition plus native tracking; this does not demonstrate phone real-time performance.

Cached gallery; predecoded query; complete warm call and image-post rendering. Retrieval-only excludes geometry/rendering.

Executed ONNX and INT8 Export

Descriptor	MB	Median ms	View R@1
FP32	45.75	24.04	41.7
INT8	11.52	17.32	33.3

- Actual float ONNX parity and QDQ INT8 inference were tested.
- Calibration uses reserved training places.
- File size and descriptor latency improve; viewpoint accuracy decreases.
- Export covers the global descriptor only.

View R@1 is the Tokyo direction-change split. No full local-matcher, CoreML or phone export result is claimed.

Limitations and Threats to Validity

- Small selected galleries and repeated physical sites.
- Synthetic night/fog cannot replace natural condition changes.
- No calibrated 6-DoF map/pose accuracy or real seasonal benchmark.
- Planar homography does not establish arbitrary 3D persistence.
- CPU timings do not establish phone power, thermal or accelerator behaviour.

Ethics and a Persistent-AR Roadmap

- Minimise camera and location retention; respect imagery terms.
- Support consent, content visibility, reporting and deletion.
- Add multi-direction references and real seasonal benchmarks.
- Introduce calibrated maps/PnP and native AR tracking integration.
- Validate wrong-place anchors and recovery on real devices.

Conclusions

- Learned local matching improves the measured illumination subset.
- Retrieval reduces learned search cost but can limit viewpoint recall.
- Short adaptation does not consistently improve held-out retrieval.
- Quantisation yields a concrete compact descriptor with an accuracy trade-off.
- Reproducible code, measured results, report and demo are delivered.

All numerical claims come from the retained Stitch experiment files. Full commands, seeds, hashes and failure analysis are in the report.

References: Features and Retrieval

- D. G. Lowe. Distinctive Image Features from Scale-Invariant Keypoints. *IJCV*, 2004.
- D. DeTone, T. Malisiewicz and A. Rabinovich. SuperPoint: Self-Supervised Interest Point Detection and Description. *CVPR Workshops*, 2018.
- P. Lindenberger, P.-E. Sarlin and M. Pollefeys. LightGlue: Local Feature Matching at Light Speed. *ICCV*, 2023.
- G. Berton, C. Masone and B. Caputo. Rethinking Visual Geo-Localization for Large-Scale Applications. *CVPR*, 2022.

References: Benchmarks and Geometry

- V. Balntas, K. Lenc, A. Vedaldi and K. Mikolajczyk. HPatches: A Benchmark and Evaluation of Handcrafted and Learned Local Descriptors. *CVPR*, 2017.
- A. Torii, R. Arandjelović, J. Sivic, M. Okutomi and T. Pajdla. 24/7 Place Recognition by View Synthesis. *CVPR*, 2015.
- P.-E. Sarlin, C. Cadena, R. Siegwart and M. Dymczyk. From Coarse to Fine: Robust Hierarchical Localization at Large Scale. *CVPR*, 2019.

Complete verified bibliography and official source URLs accompany the report.

THANK YOU

PRONOB BHATTACHARJEE

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Questions and Discussion